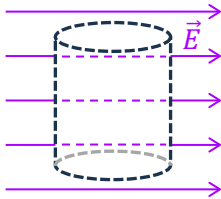


Lab 7: Electric Flux and Charge

Name: _____ Lab Partner(s): _____

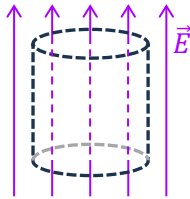
1. Each cylinder below is an imaginary closed surface. Write for each one whether the electric flux Φ through the top surface, bottom surface, curved sides, *and* the net flux Φ_{net} are positive (“+”), negative (“−”) or zero (“0”).

a.



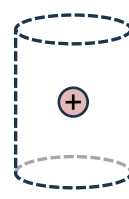
$\Phi_{\text{top}} = \underline{\hspace{2cm}}$
 $\Phi_{\text{bot}} = \underline{\hspace{2cm}}$
 $\Phi_{\text{sides}} = \underline{\hspace{2cm}}$
 $\Phi_{\text{net}} = \underline{\hspace{2cm}}$

b.



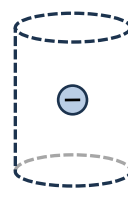
$\Phi_{\text{top}} = \underline{\hspace{2cm}}$
 $\Phi_{\text{bot}} = \underline{\hspace{2cm}}$
 $\Phi_{\text{sides}} = \underline{\hspace{2cm}}$
 $\Phi_{\text{net}} = \underline{\hspace{2cm}}$

c.



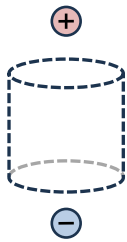
$\Phi_{\text{top}} = \underline{\hspace{2cm}}$
 $\Phi_{\text{bot}} = \underline{\hspace{2cm}}$
 $\Phi_{\text{sides}} = \underline{\hspace{2cm}}$
 $\Phi_{\text{net}} = \underline{\hspace{2cm}}$

d.



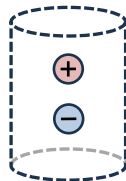
$\Phi_{\text{top}} = \underline{\hspace{2cm}}$
 $\Phi_{\text{bot}} = \underline{\hspace{2cm}}$
 $\Phi_{\text{sides}} = \underline{\hspace{2cm}}$
 $\Phi_{\text{net}} = \underline{\hspace{2cm}}$

e.



$\Phi_{\text{top}} = \underline{\hspace{2cm}}$
 $\Phi_{\text{bot}} = \underline{\hspace{2cm}}$
 $\Phi_{\text{sides}} = \underline{\hspace{2cm}}$
 $\Phi_{\text{net}} = \underline{\hspace{2cm}}$

f.



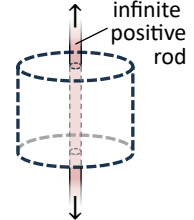
$\Phi_{\text{top}} = \underline{\hspace{2cm}}$
 $\Phi_{\text{bot}} = \underline{\hspace{2cm}}$
 $\Phi_{\text{sides}} = \underline{\hspace{2cm}}$
 $\Phi_{\text{net}} = \underline{\hspace{2cm}}$

g.



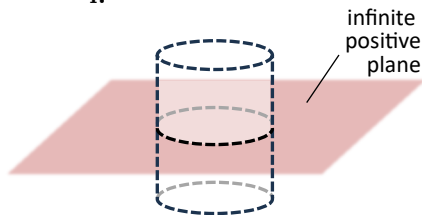
$\Phi_{\text{top}} = \underline{\hspace{2cm}}$
 $\Phi_{\text{bot}} = \underline{\hspace{2cm}}$
 $\Phi_{\text{sides}} = \underline{\hspace{2cm}}$
 $\Phi_{\text{net}} = \underline{\hspace{2cm}}$

h.



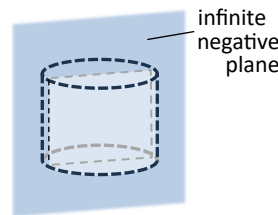
$\Phi_{\text{top}} = \underline{\hspace{2cm}}$
 $\Phi_{\text{bot}} = \underline{\hspace{2cm}}$
 $\Phi_{\text{sides}} = \underline{\hspace{2cm}}$
 $\Phi_{\text{net}} = \underline{\hspace{2cm}}$

i.



$\Phi_{\text{top}} = \underline{\hspace{2cm}}$
 $\Phi_{\text{bot}} = \underline{\hspace{2cm}}$
 $\Phi_{\text{sides}} = \underline{\hspace{2cm}}$
 $\Phi_{\text{net}} = \underline{\hspace{2cm}}$

j.



$\Phi_{\text{top}} = \underline{\hspace{2cm}}$
 $\Phi_{\text{bot}} = \underline{\hspace{2cm}}$
 $\Phi_{\text{sides}} = \underline{\hspace{2cm}}$
 $\Phi_{\text{net}} = \underline{\hspace{2cm}}$

2. Is there a relationship between the net flux Φ_{net} and how much charge is inside or outside the cylinder?